

CIS 421 Database Management Systems Final Project

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Project Plan

Objectives & Overview

BakeBase is a mini database application that aims to manage the products (baked goods), customers, orders, and ingredients of a bakery shop. The objectives of the BakeBase application are to streamline bakery operations, manage stock records, generate sales reports, and enhance management decision making.

Key Entities

Key entities of the application are listed with their respective descriptions.

Entity	Description
PRODUCTS	Manages items sold in the bakery (breads, cakes, etc.)
CUSTOMERS	Keeps track of customer information
INGREDIENTS	Monitors the bakery's stock and inventory for ingredients
ORDERS	Records orders placed by customers when purchasing items
ORDER_ITEMS	Associates products with their orders
PRODUCT_INGREDIENTS	Organizes the required ingredients for different products

Relationships Overview

1. **PRODUCTS & ORDER_ITEMS (1:N)** — Each product can appear in multiple order items, but each order item references a single product.
2. **CUSTOMERS & ORDERS (1:N)** — A customer can place multiple orders, but each order is made by a single customer.
3. **ORDERS & ORDER_ITEMS (1:N)** — Each order can have multiple items, but every order item belongs to a single order.
4. **PRODUCTS & PRODUCT_INGREDIENTS (1:N)** — Each product can require multiple ingredients, but each entry in the **PRODUCT_INGREDIENTS** table refers to a specific ingredient for a specific product.

5. **INGREDIENTS & PRODUCT_INGREDIENTS (1:N)** — Each ingredient can be used in many different products, and each entry in the **PRODUCT_INGREDIENTS** table refers to a specific ingredient for a specific product.
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Use Cases

1. **List All Orders** — Retrieve all orders in the system, including customer IDs, order dates, and due dates. Helps staff monitor/manage all active and past order history.
2. **List All Customers** — Retrieve a list of all customers in the system with their associated information including full name, phone number, and email. Ensures the bakery can contact customers for order confirmation, replacement options, payment questions, etc.
3. **View Orders and their Products** — Get a list of all orders and their associated products and quantities to provide an overview of purchases per customer. Useful for understanding what products are often purchased together, as well as provides a list for ensuring products are packaged with the correct order.
4. **List Orders by Customer** — Retrieve all orders made by customers, including their name and order dates for tracking order activity by customer name. Helps to personalize customer service and streamline order lookup.
5. **Find the Most Expensive Product** — Identify the most expensive product in the bakery to highlight the bakery's top premium product. Helps as a reference for the bakery's pricing structure.
6. **Calculate Total Amount per Order** — Calculate the total amount of each order to create sales reports and customer receipts. Aids in tracking bakery revenue and ensures accurate payment.
7. **List Customers Who Ordered Products Over a Certain Price** — Get a list of customers who ordered certain products above a specific price. Helps for targeting certain customers for reward coupons or other purchase benefits.
8. **List Customers Who Ordered a Specific Product** — Get a list of customers who ordered a specific product. This helps in tracking purchasing trends for item recommendations.
9. **Find Orders by their Total Quantity** — Identify orders where the total quantity of items exceeds a certain amount in order to prioritize larger volume orders. This helps with expediting orders.
10. **Get Top Ordered Records** — List the most ordered products by quantity sold to identify popular product trends. Assists with guiding inventory orders for ingredients and marketing popular items.